The Irreversibility Plane: Response Asymmetry and Dissipation Are Independent Coordinates of Strategic Non-Equilibrium

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Abstract

Logit revision dynamics on a finite game define a Markov jump process whose stationary state is a genuine thermodynamic object: a Gibbs equilibrium exactly when the game is potential, and a dissipative non-equilibrium steady state otherwise. Two observables are routinely treated as interchangeable readings of "how far from equilibrium" such a system sits—the Onsager asymmetry $\mathcal{R}$ of the equilibrium response matrix $\chi^{\mathrm{eq}} = (I - SB)^{-1}S$, and the Schnakenberg entropy production σ_{EP} of the revision chain. Both vanish if and only if the normalised game is potential, and unconditionally they co-move almost perfectly (Spearman $\rho_S = 0.993$, $n = 1000$). *They are not the same coordinate.* Conditioning on the harmonic fraction α of the Candogan flow decomposition, their co-movement *collapses to zero*: against that same 0.993 marginal, the within-level association falls from $\rho_S = +0.72$ at $\alpha = 0.05$ to $+0.01$ $[-0.14, +0.16]$ at $\alpha = 0.95$ at the largest action count tested ($m = 6$), intervals disjoint, and the collapse reproduces at every $m \in \{3, 4, 5, 6\}$ at $N = 2$. Independence requires decorrelation, not anti-correlation, and decorrelation is what is established. At $m \leq 5$ the residual is additionally negative ($\rho_S = -0.35$ $[-0.47, -0.23]$ at $m = 3$, $\lambda = 1.2$, against the same 0.993 marginal), but that sign reversal is a scoped fact rather than a general one—it does not survive m at fixed Frobenius payoff scale—and is quoted only with its (m, λ) attached. The obvious repair—that the collapse is an artefact of $\mathcal{R}$ being a norm ratio—was pre-registered and is refuted by its own test: the numerator $\|\chi - \chi^{\top}\|_F$ alone reaches $\rho_S = -0.368$, so no renormalisation of the response matrix recovers the dissipation ordering. This paper is not a claim that either meter is wrong. It is a claim about *dimension*: distance from equilibrium is not a scalar for strategic systems, and the $(\mathcal{R}, \sigma_{\mathrm{EP}})$ plane—not either axis—is the space in which an observed system should be classified. Both pre-registered kill-shots on that claim have now been run (5200 games, 6000 solves). *Finite size: the outcome is* ***indeterminate****, and we report why rather than around it.* Two registrations of this criterion exist. The earlier one adjudicates the raw recovery $\rho_S(6, 0.95) - \rho_S(3, 0.95)$ against bars of 0.40 (refutation) and 0.20 (survival); the later one, committed after two runs had already been launched under the earlier, adjudicates the gap ratio gap(6)/gap(3) against 0.50. On the same data the recovery is $+0.364$—resolving neither branch—while the gap ratio is 0.585, above its bar; the first registration therefore returns **indeterminate** where the second returns survival, and the deciding gap ratio's bootstrap interval crosses its own bar. We report the earlier verdict, disclose the substitution, and rest the claim elsewhere. *What the claim rests on is the ceiling criterion, and it is not close:* the high-α association never recovers at any tested action count—the worst upper interval endpoint is $+0.156$ against a 0.35 ceiling, and the $\alpha = 0.05$ and $\alpha = 0.95$ intervals are disjoint at every m. That, not a gap ratio, is what two-coordinate independence requires. *Solver:* on identical seed families a structurally different algorithm and a $100\times$ tighter tolerance perturb $\mathcal{R}$ by $\sim 10^{-12}$ and leave the rank ordering bit-identical, so the collapse is not numerical—reported as the solver-noise diagnostic it is, since a rank statistic could not have registered a perturbation that small. A registered control on the payoff scale shows the apparent finite-size decay is largely, though not wholly, a normalised-precision effect of the game generator rather than a dimensional one. We state the remaining scope honestly: the collapse is universal across every precision λ and action count m tested, but the *sign* of the residual is λ- and m-dependent, so the second dimension exists everywhere while its tilt is itself a system property; and $N = 2$ throughout, so N-scaling is untested and is the next kill-shot. We calibrate both coordinates at machine precision on systems with payoffs known by construction (Sioux Falls road network, $\mathcal{R} = 5.6 \times 10^{-17}$; Colonel Blotto, $\alpha = 0.69$, $\mathcal{R} = 0.12$), show the dissipation axis is recoverable from observed trajectories alone (rank correlation 1.0 against the exact meter; a certified finite-time thermodynamic-uncertainty lower bound at median tightness 0.85), and place two real systems in the plane—a retail scanner panel at $\mathcal{R} = 0.0011$ $[5 \times 10^{-5}, 5 \times 10^{-3}]$ and a wholesale electricity market at ≈ 1.07 nats/day—which fall in *different* quadrants. One quadrant of

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the four is not yet occupied by any measured system, and locating one there is the decisive test we name in advance. All software is open (`strataq`, Apache-2.0); every quantitative statement regenerates from fixed seeds, and Appendix A maps each number to its artifact.

Keywords: quantal response equilibrium; non-equilibrium steady state; entropy production; Onsager reciprocity; potential games; Hodge decomposition; bounded rationality

1. Introduction

Wherever a population of agents responds to incentives imperfectly, the same mathematical object appears: a softmax over payoffs. Logit quantal response equilibrium (QRE) makes that object an equilibrium concept—each agent's mixed strategy is an entropy-regularised best response to everyone else's [1, 2]—and the regularisation weight λ plays the role of an inverse temperature. The correspondence with statistical mechanics is not decorative. In an exact potential game [5], logit revision dynamics are reversible with stationary distribution $\pi^* \propto e^{\lambda\Phi}$, the probability current vanishes identically, and the entropy production is zero [6]. The system *is* a Gibbs equilibrium, not merely like one.

A *strategic non-equilibrium steady state* is what you get when that fails. Generic games are not potential; their revision dynamics break detailed balance and settle into a stationary state with circulating probability current and strictly positive entropy production [11, 12]. The question this paper is about is how one should *measure* that, given only observed behaviour.

Two measurements are natural, and both are already in use. The first is local. Perturb one agent's payoffs and observe how every agent's equilibrium mixture moves; the resulting matrix $\chi^{\text{eq}} = (I - SB)^{-1}S$ is the strategic analogue of a susceptibility, and its asymmetry is a direct test of Onsager reciprocity. The second is global. Follow the joint profile as it wanders, and measure how much the forward trajectory statistics differ from the reverse; that is the Schnakenberg entropy production of the revision chain [11]. Both vanish exactly when the normalised game is potential. Both are estimable from data without knowing payoffs—the first from cross-agent pass-through asymmetry, the second from trajectory irreversibility. It is therefore natural, and nearly universal in practice, to treat them as two views of one underlying quantity: *the* distance from equilibrium.

Prior work has not tested that assumption, because the tools to test it did not coexist. The harmonic fraction α of the Candogan flow decomposition [7] gives a basis-independent measure of how far a game is from potential, but fifteen years after its introduction there is no published software implementing it. Entropy-production estimators for observed trajectories are mature [15, 13], but they operate on scalar time series and have no representation of agents, actions or profiles. QRE solvers are excellent [21, 22] but expose no derivative of the equilibrium. Testing whether the two measurements agree requires all three at once, on the same game, at controlled α.

Scope, stated before the claim. This paper is not a new equilibrium concept, not a new solver, and not a claim that either meter is incorrect. Both are computed here exactly as their originating literatures define them; the QRE correspondence itself is validated against the reference implementation [22] to 10^{-8}. The contribution is a statement about the *dimension* of the space these meters live in, and an instrument for locating a system inside it.

The dictionary. In what follows, the "state" is a joint action profile of a finite game; the "energy" is the negative expected payoff and the "potential" is Monderer–Shapley's Φ where it exists; the "inverse temperature" is the logit precision λ; the "susceptibility" is the derivative of equilibrium play with respect to a payoff field; the "current" is the net probability flux between profiles under logit revision; and "distance from equilibrium" is precisely the thing this paper argues is not one number.

Contributions.

(C1) The two-axis result. Response asymmetry $\mathcal{R}$ and dissipation σ_{EP} share a zero and nothing else. Unconditionally they co-move at $\rho_S = 0.993$; once the harmonic fraction is conditioned on, the co-movement degrades monotonically and *decorrelates*—from $\rho_S = +0.72$ at $\alpha = 0.05$ to $+0.01$ $[-0.14, +0.16]$ at $\alpha = 0.95$ at the largest action count tested, intervals disjoint, against that same 0.993 marginal. Decorrelation, not anti-correlation, is what independence requires and what is claimed here; the negative residual seen at $m \leq 5$ is reported with its (m, λ) scope attached and is not claimed generally. The unconditional agreement is a confound: both meters are reading α ($\rho_S = 0.990$ and 0.982 respectively).

(C2) A refuted repair, pre-registered. The natural explanation— that $\mathcal{R}$ is a norm ratio whose denominator contaminates it—was registered as a hypothesis before the test and is false. The numerator alone also anti-correlates with dissipation at high α ($\rho_S = -0.368$). The mechanism is identified ($\mathcal{R}$'s within-level variation becomes denominator-driven, $\rho_S = 0.955$) but the mechanism does not repair the decoupling.

(C3) A spectral frontier, and criticality escape. A critical precision $\lambda_c(\alpha)$ exists only above $\alpha \approx 0.55$ and descends monotonically thereafter; potential games *escape* criticality at high λ because equilibrium concentration suppresses the choice covariance faster than λ amplifies it.

(C4) Observability. The dissipation axis is recoverable from observed trajectories alone—a k-block KLD estimator tracks the exact meter at rank correlation 1.0, and a debiased finite-time thermodynamic-uncertainty bound [14] certifies a lower bound at median tightness 0.85—and both axes are calibrated at machine precision against systems whose payoffs are known by construction.

(C5) Software and reproducibility. `strataq` [27], an open JAX library, is the first published implementation of the Candogan decomposition, of χ^{eq} for logit QRE, and of agent-aware entropy production. Every figure and number in this paper regenerates from committed artifacts and fixed seeds.

To the best of the author's knowledge this is the first statement that the local and global irreversibility measures of a strategic system are structurally inequivalent. No claim is made that either meter supersedes the other, and no claim is made about the sign of the residual correlation: Section 5. shows that sign to be λ- and m-dependent, and Section 6. treats it as an open question rather than a result.

Hypotheses. Registered before the runs reported here, with the test and the level fixed in advance (criteria files are committed and their commits verified landed; see Appendix A).

H1 Within-α-level $\rho_S(\sigma_{\mathrm{EP}}, \mathcal{R})$ declines as $\alpha \to 1$. *Test:* Spearman within each of ten α levels, 100 games per level; monotone decline over the top three levels. *Primary.*

H2a The decline is a ratio artefact: $\|\chi - \chi^\top\|_F$ alone retains co-movement with σ_{EP} at high α ($\rho_S > 0.5$).

H2b At high α, $\mathcal{R}$'s within-level variation is denominator-driven: $\rho_S(\mathcal{R}, 1/\|\chi + \chi^\top\|_F) > 0.8$.

H3 The collapse is universal in $\lambda \in \{0.8, 1.2, 2.0\}$ and $m \in \{3, 4\}$; **H3b**: its residual sign is λ-invariant.

H4 Both coordinates are estimable on real systems without payoff knowledge.

H5 *Kill-shot, finite size.* The collapse is an artefact of $m = 3$ and recovers as the action set grows. *Test:* within-level ρ_S over $m \in \{3, 4, 5, 6\}$, 200 games per (m, α) cell, $N = 2$, $\lambda = 1.2$, percentile bootstrap (2000 resamples) on every cell. Writing $\mathrm{gap}(m) = \rho_S(m, 0.05) - \rho_S(m, 0.95)$, the claim *dies* iff all three of (a) $\rho_S(m, 0.95)$ is monotone non-decreasing in m to within 0.05, (b) $\mathrm{gap}(6)/\mathrm{gap}(3) < 0.50$, and (c) the terminal intervals at $m = 3$ and $m = 6$ are disjoint; it *survives* iff all three of $\mathrm{gap}(m)/\mathrm{gap}(3) \geq 0.50$, $\rho_S(m, 0.95) \leq 0.35$, and the upper interval endpoint below 0.35, at every m. Anything else is **indeterminate**—the branches are deliberately not exhaustive, so the run cannot pass by default. Precondition: $\rho_S(m, 0.05) \geq 0.55$ at every m. *Primary.*

H5b *Onset.* The α at which the collapse begins (smallest swept level with $\rho_S < 0.5$) does not drift upward with m by more than 0.10, one grid step.

H6 *Kill-shot, solver.* The collapse is an artefact of the damped fixed-point solver's finite-precision σ^*. *Test:* recompute the $\alpha \in \{0.85, 0.95\}$ cells at $m = 3$ on the identical seed family with magnetic mirror descent (a structurally different algorithm) and with the damped solver at a $100\times$ tighter tolerance; the collapse is numerical if $|\Delta\rho_S| \geq 0.05$ in any of the four comparisons. σ_{EP} is built from the generator and never touches the solver, so the solver can enter this correlation only through $\mathcal{R}$.

H5 and H6 were registered in a configuration file committed *before* the experiment implementing them existed. H5 was registered *twice*: the criterion stated above is the second registration, and an earlier one—adjudicating the raw recovery $\rho_S(6, 0.95) - \rho_S(3, 0.95)$ against bars of 0.40 and 0.20—was live when the first two runs of this sweep were launched. The substitution, the disagreement between the two on the same data, and the verdict we report as a result are set out in Section 5.8. and among the limitations in Section 6.. The outcome—**indeterminate** on H5 with the claim resting on the ceiling condition instead, a solver-noise diagnostic rather than a survived kill-shot on H6, and a confound found along the way—is carried in the scoreboard (Table 5) with the same wording as the hypotheses that failed.

Section 2. positions the work; Section 3. defines the two coordinates; Section 4. gives the game families, seeds and statistics; Section 5. reports the results and the hypothesis scoreboard; Section 6. states the mechanism, what $\mathcal{R}$ is not, and the limitations; Section 7. concludes.

2. Background and Related Work

2.1. Quantal response and its empirical content

Logit QRE [1] replaces exact best response with $\sigma_i(a) \propto \exp\{\lambda_i U_i(a;\sigma_{-i})\}$, a fixed point that interpolates between uniform play ($\lambda \to 0$) and Nash ($\lambda \to \infty$). It is exactly the argmax of expected payoff plus λ^{-1} times Shannon entropy [17], and λ admits a rational-inattention reading as the inverse shadow price of information [16]. Its principal empirical threat is Haile, Hortaçsu and Kosenok [3]: unrestricted quantal response rationalises any behaviour. The standard defences are the regular-QRE axioms [2] and external identification of payoffs. A third defence is used here and is structural rather than parametric: the reciprocity test of Section 3. constrains the *symmetry* of the response matrix and admits no free parameter to absorb misfit.

2.2. Statistical mechanics of games

In potential games logit dynamics are reversible with a Gibbs stationary measure [6, 5]; Brock and Durlauf's social-interactions model [18] is the mean-field Ising analogue and supplies the multiplicity transition. Away from potential games the correspondence is not merely approximate—it changes character. The chain sustains current and dissipates, and the machinery of stochastic thermodynamics applies: Schnakenberg's cycle decomposition [11], fluctuation theorems, and the thermodynamic uncertainty relation [14, 12]. Quantal response statistical equilibrium [19] is the closest existing econophysics programme; it is a constrained-MaxEnt model over aggregate outcomes and, by its authors' own account, cannot represent out-of-equilibrium dynamics—which is precisely the regime studied here.

2.3. Game decomposition

Candogan et al. [7] decompose a finite game orthogonally into potential, harmonic and nonstrategic components, giving a basis-independent harmonic fraction α. Sandholm's externality symmetry [8] and the Jacobian splitting of Balduzzi et al. [9] characterise potential games; the full-versus-effective distinction matters, and all decomposition here is taken on the normalised (strategically equivalent) game. Legacci, Mertikopoulos and Pradelski [10] study convergence versus recurrence under exponential weights across the same potential–harmonic axis, in continuous-time replicator geometry; the present work is discrete-time, works with response matrices and stationary fluxes, and is complementary rather than competing.

2.4. Fluctuation and response in economic settings

Garnier-Brun, Bouchaud and Benzaquen [20] establish a fluctuation–response identity for the consumer-side Slutsky matrix, which is the demand-side counterpart of the exact static identity $\chi^{\mathrm{part}} = \lambda C$ used here. What that literature has not addressed is the *equilibrium* response—the object obtained after strategic feedback—or its relationship to dissipation.

2.5. Gap analysis

Every ingredient exists; no tool computes them together. Gambit [22] is authoritative for the QRE correspondence but exposes no derivative of the equilibrium, no decomposition, and no dynamics. Entropy-production estimators are mature but operate on scalar series and cannot attribute dissipation to an agent. The Candogan decomposition has no published implementation of any kind. **The shared defect is that no existing tool can compute a local response object and a global dissipation object for the same game at a controlled harmonic fraction**—which is exactly the comparison this paper makes. Table 1 states the position.

Table 1. Capability matrix. Rows are existing tools; columns are the operations required to compare a local response object with a global dissipation object on one game.

Tool	**QRE**	$d\sigma^*/du$	**Decomp.** α	**Dynamics,** σ_{EP}	**Agent-aware**
Gambit / pygambit [22]	Yes	No	No	No	—
nashpy	No	No	No	No	—
OpenSpiel (MMD) [23]	Partial	No	No	No	—
QuantEcon (`logitdyn`)	No	No	No	Partial	Yes
Scalar irreversibility tools [15]	No	No	No	Yes	**No**
`strataq` [27]	Yes	Yes	Yes	Yes	Yes

3. The Two Coordinates

3.1. Notation

Players $i \in \{1, \ldots, N\}$ with finite action sets A_i, $|A_i| = m_i$; mixed strategies $\sigma_i \in \Delta(A_i)$ with full support. Expected payoffs $U_i(a; \sigma_{-i}) = \sum_{a_{-i}} \big(\prod_{j \neq i} \sigma_j(a_j)\big) u_i(a, a_{-i})$, and logit QRE

$$\sigma_i(a) = \frac{\exp\{\lambda_i U_i(a; \sigma_{-i})\}}{\sum_b \exp\{\lambda_i U_i(b; \sigma_{-i})\}}. \tag{1}$$

Write $C_i := \operatorname{diag}(\sigma_i) - \sigma_i \sigma_i^\top$ for the choice covariance, $S := \operatorname{blockdiag}(\lambda_i C_i)$, and $B_{ij}(a, b) := \partial U_i(a; \sigma_{-i})/\partial \sigma_j(b)$ for $j \neq i$, $B_{ii} := 0$. C_i is rank-deficient by construction, so all linear algebra below is performed on the tangent space $T = \bigoplus_i \{v : \mathbf{1}^\top v = 0\}$ in an explicit Helmert basis; omitting that projection fabricates a zero eigenvalue and a spurious criticality reading.

3.2. The local coordinate

The exact static identity $\chi_i^{\text{part}} = \partial\sigma_i/\partial h_i|_{\sigma_{-i}} = \lambda_i C_i$ is a fluctuation–dissipation relation with opponents frozen. Total differentiation of Equation (1) under a payoff field h gives $d\sigma = S(dh + B\, d\sigma)$, hence the *strategic resolvent*

$$\chi^{\text{eq}} := \frac{d\sigma^*}{dh} = (I - SB)^{-1} S. \tag{2}$$

χ^{eq} is symmetric if and only if $S(B - B^\top)S = 0$, which under full support and on T is equivalent to $B = B^\top$, i.e. to the normalised game having zero harmonic component. Strategic feedback therefore neither creates nor destroys reciprocity, and the departure is observable. Define the *response asymmetry*

$$\mathcal{R} := \frac{\|\chi^{\text{eq}} - (\chi^{\text{eq}})^\top\|_F}{\|\chi^{\text{eq}} + (\chi^{\text{eq}})^\top\|_F}. \tag{3}$$

$\mathcal{R}$ is estimable from cross-agent pass-through asymmetry—how much agent i moves when agent j's cost moves, against the reverse—without knowing payoffs. Its *zero test* is λ-free; its magnitude is not, and reporting a magnitude without the λ it was read at is an error.

3.3. The global coordinate

Logit revision defines a Glauber jump process on the joint profile space with $N_{\text{states}} = \prod_i m_i$. At stationarity the pairwise current is $J^*(a, a') = \pi^*(a) w(a \to a') - \pi^*(a') w(a' \to a)$ and the entropy production rate is

$$\sigma_{\text{EP}} = \tfrac{1}{2} \sum_{a, a'} \big[\pi^*(a) w(a \to a') - \pi^*(a') w(a' \to a)\big] \log \frac{\pi^*(a) w(a \to a')}{\pi^*(a') w(a' \to a)} \;\geq\; 0, \tag{4}$$

non-negative by construction and zero exactly under detailed balance. From observed trajectories alone we use a k-block KLD estimator and a debiased finite-time thermodynamic-uncertainty lower bound [14]; the latter is a certified bound rather than a point estimate and degrades gracefully under partial observation.

3.4. The confound, and why it must be conditioned away

Both coordinates are monotone in the harmonic fraction $\alpha(u) := \|u^H\|/(\|u^P\| + \|u^H\|)$ of the normalised game's Candogan decomposition [7], computed here by a separable Kronecker transform in near-linear time. Any unconditional correlation between $\mathcal{R}$ and σ_{EP} is therefore substantially a correlation of each with α. **The entire empirical design of this paper is the decision to stratify on α before comparing them.**

3.5. The plane

Equations (3) and (4) define a map from a game (or an observed system) to a point in the quadrant $[0, \infty)^2$. Figure 1 shows the two computational routes; they share an origin and are otherwise unrelated in type—one is a derivative evaluated at a single equilibrium, the other a functional of a flux field over the whole profile space.

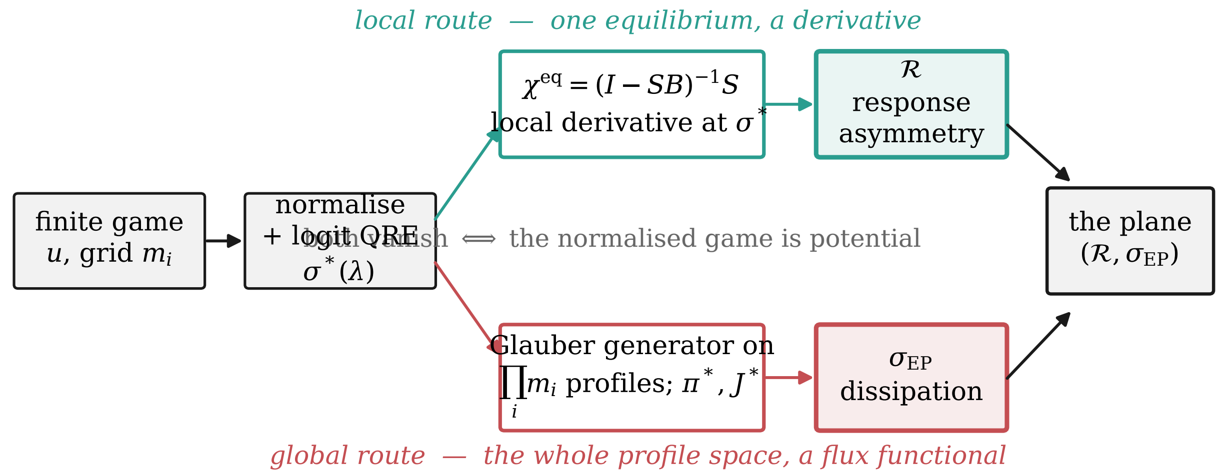

Figure 1. The two routes from a finite game to a coordinate. Both terminate in zero exactly when the normalised game is potential; nothing else forces them to agree.

4. Experimental Setup

4.1. Game families

Synthetic families are generated by projecting a seed game onto its potential and harmonic components and re-mixing at an exact target α, never by perturbing a Jacobian (the skew part of a Jacobian is coordinate-dependent and generally not integrable). Ten levels $\alpha \in \{0.05, 0.15, \ldots, 0.95\}$, 100 games per level, $N = 2$, $m = 3$ unless stated. Calibration anchors have payoffs known by construction: Rosenthal congestion on the real Sioux Falls network (an exact potential game) and Colonel Blotto (free payoffs, high α).

4.2. Solvers and numerics

Damped logit fixed-point iteration with all softmax evaluations via `log_softmax`; `float64` enforced globally, which is required because $(I - SB)$ is ill-conditioned near criticality. Cross-validated against the reference homotopy implementation [22] to 10^{-8} on 2×2 and 3×3 games, and χ^{eq} against finite differences of the solved equilibrium to 1.3×10^{-8}.

4.3. Metrics

$\mathcal{R}$ from Equation (3); σ_{EP} from Equation (4) on the dense Glauber generator with π^* obtained by SVD nullspace; α from the separable Kronecker Hodge transform; $\rho(SB)$ and its bifurcation classification from the spectrum of SB on the tangent space.

4.4. Statistical analysis

Association is measured by Spearman ρ_S throughout, because neither meter is expected to relate linearly to the other. Confidence intervals are bootstrap percentile (2000 resamples) for synthetic families and cluster bootstrap over stores (500 resamples) for panel data. Sample sizes are justified in advance in the criteria files: $n = 1000$ (100 games $\times$ 10 levels) puts the bootstrap CI half-width on ρ_S below 0.02, an order of magnitude inside the registered thresholds.

4.5. Nulls

No detection verdict is reported against a single null. For trajectory irreversibility the reported verdict requires exceeding both a Fourier-transform surrogate and an amplitude-adjusted FT surrogate null, with a reversibilized-Markov null as the decisive comparison; a value-space reading serves as a built-in blindness control, because a periodic drive retraces its one-dimensional path and must therefore sit at null in value space.

4.6. Pre-registration and reproducibility

Criteria are written into a committed configuration file *before* each run and the commit is verified landed. Every run writes a resolved-configuration snapshot next to its artifact, and each artifact records its metrics, effect sizes with confidence intervals, n with a justification, and its seed. Editing a criterion after seeing results is prohibited; where a registered criterion failed, that fact is reported rather than the criterion revised (Section 5.). All artifacts and the figure script are public; `make reproduce` regenerates every number from its seed.

5. Results

5.1. The unconditional agreement, and why it is a confound

Across 1000 game–level pairs the two meters agree almost perfectly: $\rho_S(\sigma_{\mathrm{EP}}, \mathcal{R}) = 0.9927$ $[0.9914, 0.9936]$. Taken alone this is the result a less careful design would report. It is a confound. Each meter is separately near-perfectly monotone in the harmonic fraction: $\rho_S(\sigma_{\mathrm{EP}}, \alpha) = 0.9904$ and $\rho_S(\mathcal{R}, \alpha) = 0.9819$ ($n = 2000$). The unconditional correlation is therefore substantially the correlation of two functions of α with each other, and says nothing about whether they carry independent information.

5.2. Conditional on α, the co-movement collapses

Stratifying by α level (Table 2, Figure 2a) the within-level association is strong and flat for $\alpha \leq 0.65$, degrades over the top three levels, and at this action count crosses zero into negative territory at $\alpha = 0.95$. The crossing is scoped: Section 5.8. shows that the *collapse* reproduces at every action count tested while the *negative sign* does not, so it is the collapse and not the reversal that carries the two-coordinate claim.

Table 2. Within-α-level Spearman association between dissipation and response asymmetry. 100 games per level, $\lambda = 1.2$, $N = 2$, $m = 3$.

α	0.05	0.15	0.25	0.35	0.45	0.55	0.65	0.75	0.85	0.95
$\rho_S(\sigma_{\mathrm{EP}}, \mathcal{R})$	0.882	0.812	0.856	0.849	0.800	0.870	0.801	0.610	0.323	−0.355
$\rho_S(\sigma_{\mathrm{EP}}, \Vert\chi - \chi^\top\Vert_F)$	0.768	0.749	0.823	0.808	0.757	0.857	0.792	—	—	−0.368

H1 is accepted. The decline over the top three levels is monotone and the terminal value is negative at $m = 3$; the decline itself is confirmed at every m tested in Section 5.8..

5.3. The repair hypothesis is refuted by its own test

$\mathcal{R}$ is a ratio of norms, so the natural explanation of the collapse is that the denominator $\Vert\chi + \chi^\top\Vert_F$—a measure of the *symmetric* response, which grows with α—contaminates the numerator's signal. This was registered as H2a before the test: if true, the numerator alone should retain co-movement with σ_{EP} at high α.

It does not. The numerator alone reaches $\rho_S = -0.368$ at $\alpha = 0.95$ (Figure 2a, dashed red), tracking the ratio almost exactly through the collapse. **H2a is not supported**, and the consequence is the paper's central claim: *no renormalisation of the response matrix recovers the dissipation ordering at high* α.

H2b—that $\mathcal{R}$'s within-level variation becomes denominator-driven—**is accepted**: $\rho_S(\mathcal{R}, 1/\|\chi+\chi^\top\|_F)$ rises monotonically from -0.082 at $\alpha = 0.05$ to 0.955 at $\alpha = 0.95$ (Figure 2b). So the mechanism by which $\mathcal{R}$ loses contact with dissipation is identified—at high α the ratio is measuring residual symmetric response rather than asymmetry—but identifying it does not repair it, because the numerator has lost contact too.

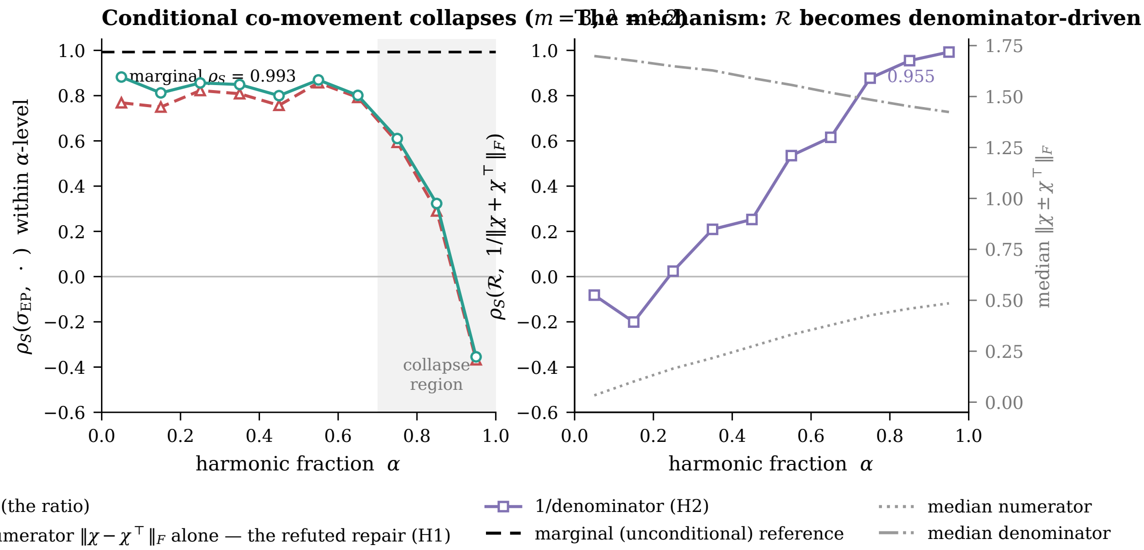

Figure 2. (**a**) Within-level association between dissipation and response asymmetry against harmonic fraction, with the unconditional value as a reference. (**b**) The mechanism, and the two norms that produce it. The refuted repair hypothesis is the dashed red series in (**a**); by construction it would have to lie above the teal series for H2a to hold.

5.4. The plane, and where measured systems sit in it

Figure 3 plots both coordinates together. The synthetic α-family runs up the diagonal, which is what a one-dimensional theory of distance-from-equilibrium predicts and is the reason the confound is so persuasive. The measured systems do not lie on it. A retail scanner panel reads $\mathcal{R} = 0.0011$ $[4.8 \times 10^{-5}, 5.0 \times 10^{-3}]$ ($n = 22{,}655$ store-weeks over 86 stores, cluster bootstrap)—potential-like, exactly where a single-retailer category-management model predicts, a prediction registered before the run. A wholesale electricity day-ahead market reads ≈ 1.07 nats/day against a reversibilized-Markov null ($p < 0.01$) while sitting at null against Fourier and amplitude-adjusted surrogates. **Those two readings fall in different quadrants**, and each currently has only one coordinate measured—which the figure shows as a band rather than a point.

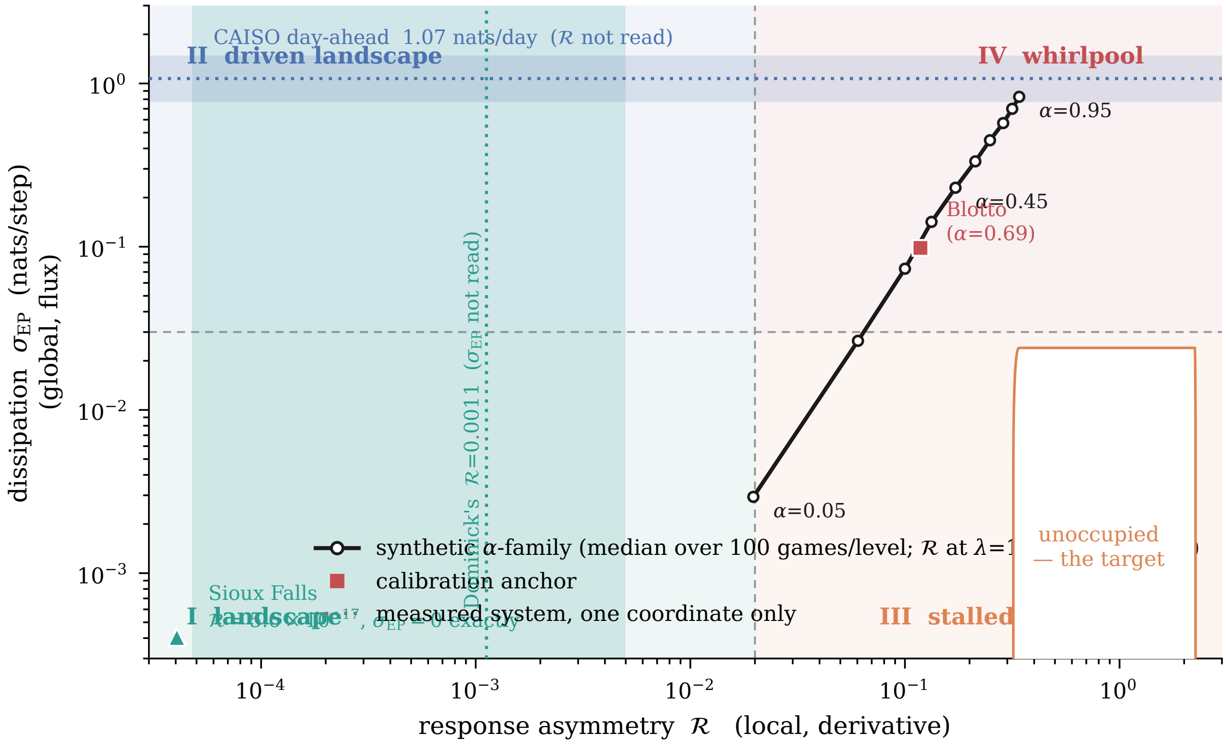

Figure 3. The irreversibility plane. Quadrant shading uses the library's calibrated bands. Bands rather than points indicate a system for which only one coordinate has been read.

5.5. Criticality escape and the frontier

The same operator $(I - SB)$ that gives the response also certifies uniqueness, and its spectral radius locates the transition. On potential games $\rho(SB)$ is *non-monotone* in λ: it rises, peaks, and falls, because equilibrium concentration kills the choice covariance faster than λ amplifies it. Potential games therefore escape criticality at high precision. A crossing of $\rho(SB) = 1$ exists only for $\alpha \geq 0.55$, and above that the critical precision descends monotonically—$\lambda_c = 7.77, 5.22, 4.22, 3.64, 3.23, 2.95$ for $\alpha = 0.55$ to 0.80, single crossing verified at every level, zero monotonicity violations, $n = 440$ (Figure 4).

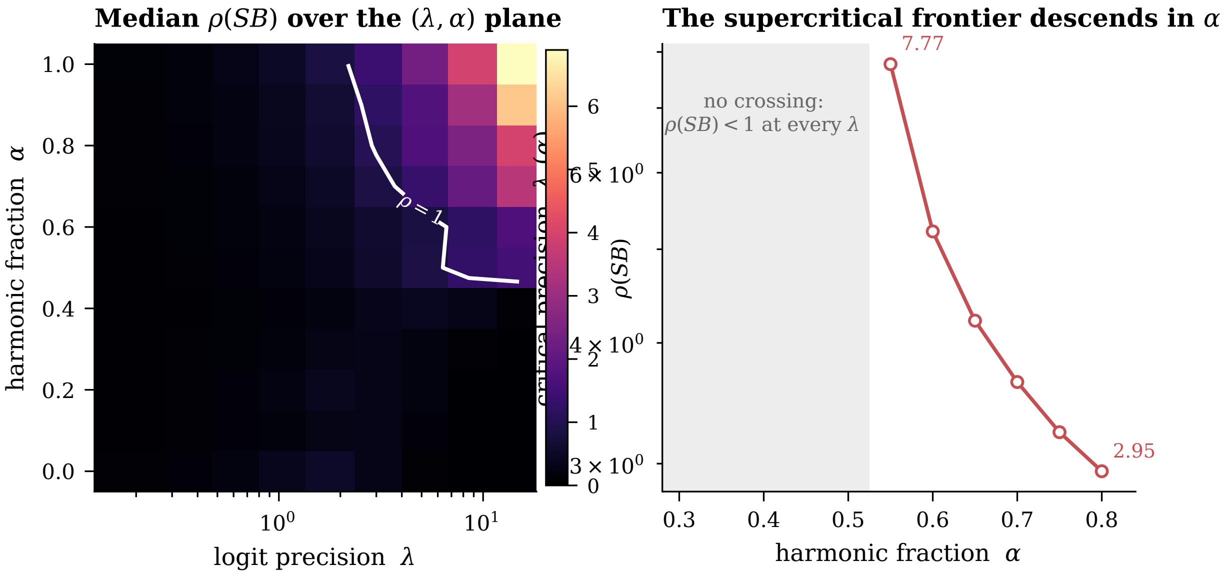

Figure 4. (**a**) Median spectral radius of SB over the precision–harmonicity plane, with the unit contour. (**b**) The critical precision, which exists only above $\alpha \approx 0.55$.

5.6. The dissipation axis is observable

Against the exact meter, the k-block KLD estimator attains rank correlation 1.0 across the α sweep, and the debiased finite-time thermodynamic-uncertainty bound lies below the exact value at every level with median tightness 0.850 (Figure 5). The bound, not the point estimate, is the reportable statement on real data.

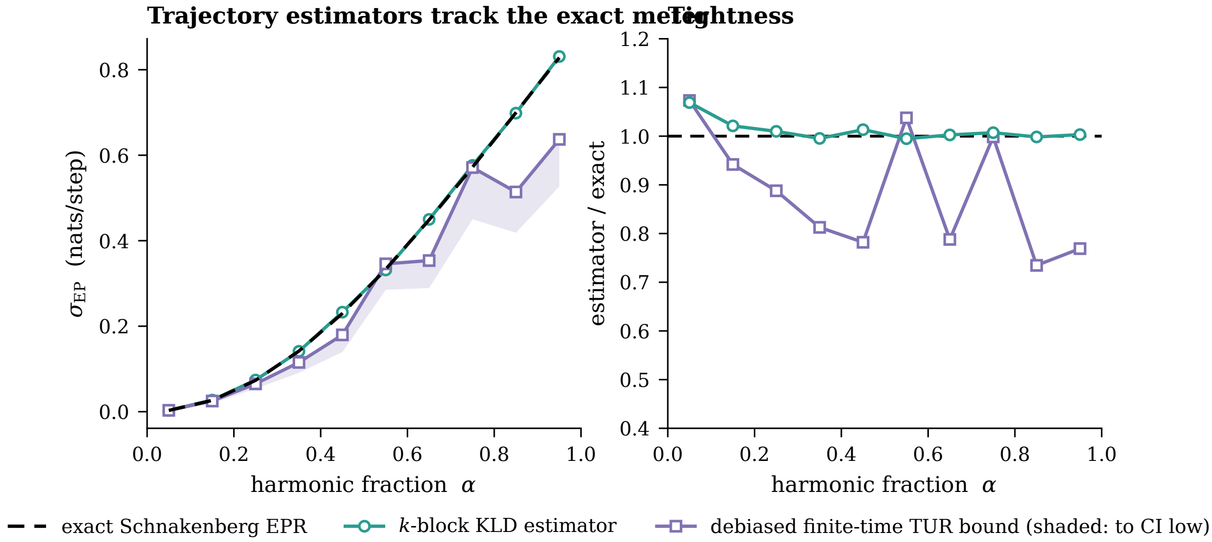

Figure 5. (**a**) Trajectory estimators against the exact entropy production. (**b**) Tightness. By construction the certified bound cannot exceed the exact value.

5.7. Sensitivity: what is universal and what is not

Repeating the stratified sweep across $\lambda \in \{0.8, 1.2, 2.0\}$ and $m \in \{3, 4\}$ separates two things that a single run conflates. **The collapse is universal**: the conditional association degrades at every setting tested. **The sign of the residual is not**: it runs $+0.25 \to +0.03 \to -0.23$ as λ increases, and Section 5.8. adds m to the list of things it varies with. **H3 is accepted; H3b is not supported.** We therefore report the existence of a second dimension as the result, and the direction of its tilt as a system-specific property and an open question. This distinction was not anticipated; a registered criterion failed 2 of 4 cells and was revised on the record rather than quietly, and a follow-on "wedge-depth" explanation for the reversal was refuted (0 of 180 games supercritical).

5.8. Two pre-registered kill-shots

The two most plausible ways the two-coordinate claim is wrong are that the collapse is small-system noise that recovers as the action set grows (H5), and that it is an artefact of the fixed-point solver (H6). Both were registered with their thresholds in a configuration file committed before the experiment implementing them existed, and both have been run: 5200 distinct games (6000 solves) at $N = 2$, $\lambda = 1.2$, 200 games per main cell, percentile bootstrap (2000 resamples) on every reported ρ_S. The seed stream was reconstructed from the reference run, so the first 100 games of every $m = 3$ cell are literally the games behind Table 2; the replication deviation at $\alpha \in \{0.85, 0.95\}$ is 0.000, which makes every comparison below anchored to the published reading rather than to a fresh sample of it.

Table 3. Finite-size sweep (H5). Within-level Spearman association at each action count, 200 games per cell, $N = 2$, $\lambda = 1.2$, fixed Frobenius payoff scale. Brackets are 95% percentile-bootstrap intervals; gap $= \rho_S(0.05) - \rho_S(0.95)$.

m	α=0.05	0.45	0.75	0.85	α=0.95 [95% CI]	gap	gap ratio
3	+0.851	+0.869	+0.627	+0.398	−0.352 [−0.473, −0.226]	1.203	1.000
4	+0.806	+0.773	+0.685	+0.438	−0.272 [−0.405, −0.140]	1.078	0.896
5	+0.702	+0.741	+0.654	+0.439	−0.244 [−0.377, −0.097]	0.947	0.787
6	+0.716	+0.698	+0.607	+0.388	+0.012 [−0.137, +0.156]	0.704	0.585

H5 is indeterminate, and the reason is a criterion substitution we report on the record. This unit has *two* registrations of the same kill-shot. The first (`plane_finite_size.yaml`, committed 23:02:25) adjudicates the raw recovery $\Delta\rho_{\rm hi} = \rho_S(6, 0.95) - \rho_S(3, 0.95)$: refutation iff $\Delta\rho_{\rm hi} > 0.40$, survival iff $\Delta\rho_{\rm hi} \leq 0.20$. An experiment implementing it was written at 23:04:39, executed at 23:04:49, and run again at 23:17:47. The second (`plane_robustness.yaml`, committed 23:26:36—*after* both of those runs) replaces the deciding statistic with the gap ratio gap(6)/gap(3) against a 0.50 bar. On the data of Table 3 the two registrations disagree, and the change ran in the direction of survival: $\Delta\rho_{\rm hi} = +0.364$ is neither above 0.40 nor at or below 0.20, so the first registration returns **indeterminate**, while the gap ratio is 0.585, above its bar, so the second returns survival. Both files define *indeterminate* as a third outcome for exactly this situation. Where two registrations of one criterion disagree and the earlier was live when the data were first produced, the earlier binds: **H5 is indeterminate**, and the claim's status under it is unchanged rather than strengthened.

The deciding statistic reached adjudication without an interval, and its interval crosses the bar. The second registration required bootstrap intervals on every reported ρ_S, and stated the principle that a point estimate on the right side of a threshold whose interval crosses it does not certify survival. The gap ratio is not a ρ_S, so it received no interval. Bootstrapped after the fact (4000 resamples, games resampled within cell, both α cells resampled per replicate, $m = 3$ as the shared denominator), gap(6)/gap(3) = 0.585 with 95% interval $[+0.434, +0.749]$ and $P(\text{gap ratio} < 0.50) = 0.129$—a margin of 1.08 bootstrap standard deviations. The interval crosses 0.50. By the registration's own principle the surviving branch is not certified, which is a second and independent route to the same verdict. Two of the three refutation conditions did fire: ρ_S at $\alpha = 0.95$ rises with m with no dip ($-0.352 \to -0.272 \to -0.244 \to +0.012$), and the terminal intervals at $m = 3$ and $m = 6$ are disjoint.

What is robust, and what the claim rests on, is the ceiling. The high-α association never recovers to coupled behaviour at any tested action count: $\rho_S(m, 0.95) \leq 0.35$ at every m, and the *upper* interval endpoint is below the ceiling at every m—worst +0.156 against 0.35. The $\alpha = 0.05$ and $\alpha = 0.95$ intervals are disjoint at every m, and the precondition $\rho_S(m, 0.05) \geq 0.55$ holds throughout. That is what two-coordinate independence requires, and unlike the gap ratio it is not a near thing. It is also *immune to the substitution*: the 0.35 ceiling and its interval form are identical in both registrations—and are themselves inherited from an earlier unit's registered collapse threshold rather than chosen here—so whichever registration binds, this condition reads the same way and passes with room. The substitution moved the primary statistic and nothing else. No indeterminacy guard fired: zero near-critical games, zero non-convergences and zero generator rejections across all 6000 solves; the verdict above is a criterion outcome, not a data-quality one. The $m = 3$ row is a re-execution of the reference cells on the identical games, deviation 0.000, which is the strongest single property of the run and is independent of every criterion dispute in it.

What does not survive is the sign. At $m = 6$ the terminal association is $+0.012$ $[-0.137, +0.156]$—statistically indistinguishable from zero. The claim this paper makes is therefore the weaker and sufficient one: *the coupling collapses to zero at high* α—at $m = 6$, from $+0.716$ to $+0.012$ with disjoint intervals—and the collapse itself is present at every action count tested. Independence requires decorrelation, not anti-correlation. The sign reversal reported at $m = 3$ in Table 2 is retained only with its $(m, \lambda_{\rm norm})$ scope attached.

H5b: the onset does not drift. The smallest swept level at which $\rho_S < 0.5$ is $\alpha = 0.85$ at every $m \in \{3, 4, 5, 6\}$—drift 0.00 against a tolerance of 0.10. Where the collapse begins is m-independent; only how far it goes at $\alpha = 0.95$ moves with m, and, as the next paragraph shows, only at fixed Frobenius scale.

5.8.1. The m-trend is largely, not wholly, a normalised-precision effect

The generator unit-normalises each Hodge component and multiplies by a fixed Frobenius scale, so per-entry payoff RMS falls like $1/m$: a larger action set is also a *colder* game. Across the main sweep at $\alpha = 0.95$ the median normalised precision $\bar{\lambda} = \lambda \cdot (\text{payoff range})$ falls $1.86 \to 1.62 \to 1.42 \to 1.27$ and the median $\sigma_{\rm EP}$ falls $0.546 \to 0.318 \to 0.206 \to 0.144$ as m goes $3 \to 6$. The registered control D1 repeats the extreme levels under the a-priori rule $\text{scale}(m) = 2.0\, m/3$, which holds per-entry RMS approximately constant ($\bar{\lambda} = 1.85 \to 2.18 \to 2.38 \to 2.58$):

Table 4. Payoff-scale control (D1), registered before the run. 100 games per cell, same levels, same λ. The control raises $\bar{\lambda}$ with m where the main arm lowers it, so the two arms *bracket* the confound rather than matching it.

m	$\rho_S(\alpha{=}0.05)$	$\rho_S(\alpha{=}0.95)$ [95% CI]	$\bar{\lambda}(\alpha{=}0.95)$	gap ratio
3	+0.852	−0.347 [−0.516, −0.170]	1.85	1.000
4	+0.625	−0.431 [−0.584, −0.251]	2.18	0.881
5	+0.779	−0.453 [−0.617, −0.258]	2.38	1.027
6	+0.747	−0.300 [−0.471, −0.116]	2.58	0.872

In the control arm the terminal association is non-monotone in m, the negative sign is intact at every m including $m = 6$, and the collapse at $m = 6$ retains 87% of its $m = 3$ size instead of 59% (Figure 6b). **This does not mean the m-trend vanishes at matched effective precision, and we withdraw that reading.** D1 does not hold $\bar{\lambda}$ constant: it drives $\bar{\lambda}$ *up* 40% across $m \in \{3, \ldots, 6\}$ ($1.85 \to 2.58$) where the main arm drives it *down* 32% ($1.86 \to 1.27$). The two arms therefore bracket the confound rather than controlling it, and the honest reading is an interpolation between them: taking $\rho_S(6, 0.95)$ linearly in $\bar{\lambda}$ between the two arms and evaluating at the $m = 3$ arms' common $\bar{\lambda} \approx 1.86$ gives ≈ -0.13, leaving a residual m-effect of ≈ 0.22 of the observed 0.364. **Most of the m-effect survives matching.** The supported statement is that the apparent finite-size decay is *largely, not wholly*, a normalised-precision effect of the game generator, consistent with this programme's earlier results on the λ-sensitivity of both meters. Both arms return the same verdict under the second registration, so the registered disagreement rule does not fire; but D1 is *unpaired* to the main arm—a different seed offset and half the games per cell—so every arm-to-arm comparison here carries between-sample noise a paired design would cancel, and the control should be re-run paired at $\bar{\lambda}$ fixed by construction. The methodological consequence stands and is restated among the limitations in Section 6.: an m-sweep on this family that does not control per-entry payoff RMS will confound temperature with dimension.

5.8.2. The collapse is not numerical, and the solver check is a diagnostic

H6 returns $|\Delta\rho_S| = 0.00000$, and that number had no way to come out otherwise. At $m = 3$, on the identical 100-game families at $\alpha \in \{0.85, 0.95\}$, magnetic mirror descent and the damped solver at a 100× tighter tolerance both reproduce the reference association exactly, in all four comparisons, against a registered bar of 0.05. The supporting diagnostics say why: the maximum sup-norm deviation in the equilibrium mixture is 2.7×10^{-12} and the maximum deviation in $\mathcal{R}$ itself is 1.3×10^{-12}. ρ_S is a *rank* statistic, and a perturbation twelve orders of magnitude below the spacing of adjacent $\mathcal{R}$ values in a 100-game sample cannot reorder them; $|\Delta\rho_S| = 0$ was therefore arithmetically forced before the run rather than discovered by it, and its "margin" against the 0.05 bar is not evidence. We report H6 as what it actually measures—**the solver contributes $\sim 10^{-12}$ to $\mathcal{R}$, and the rank ordering of $\mathcal{R}$ is bit-stable under solver replacement**—which is a real result and excludes finite-precision path dependence, the stated H6 hypothesis. It is not a kill-shot passed with margin. A test with power would have to perturb $\mathcal{R}$ at the scale of the rank spacing, or use a statistic sensitive to magnitude. Since σ_{EP} is computed from the generator and never touches the solver, the solver can enter this correlation only through $\mathcal{R}$; and no comparison of two methods converging to the same fixed point can exclude a bias shared by both.

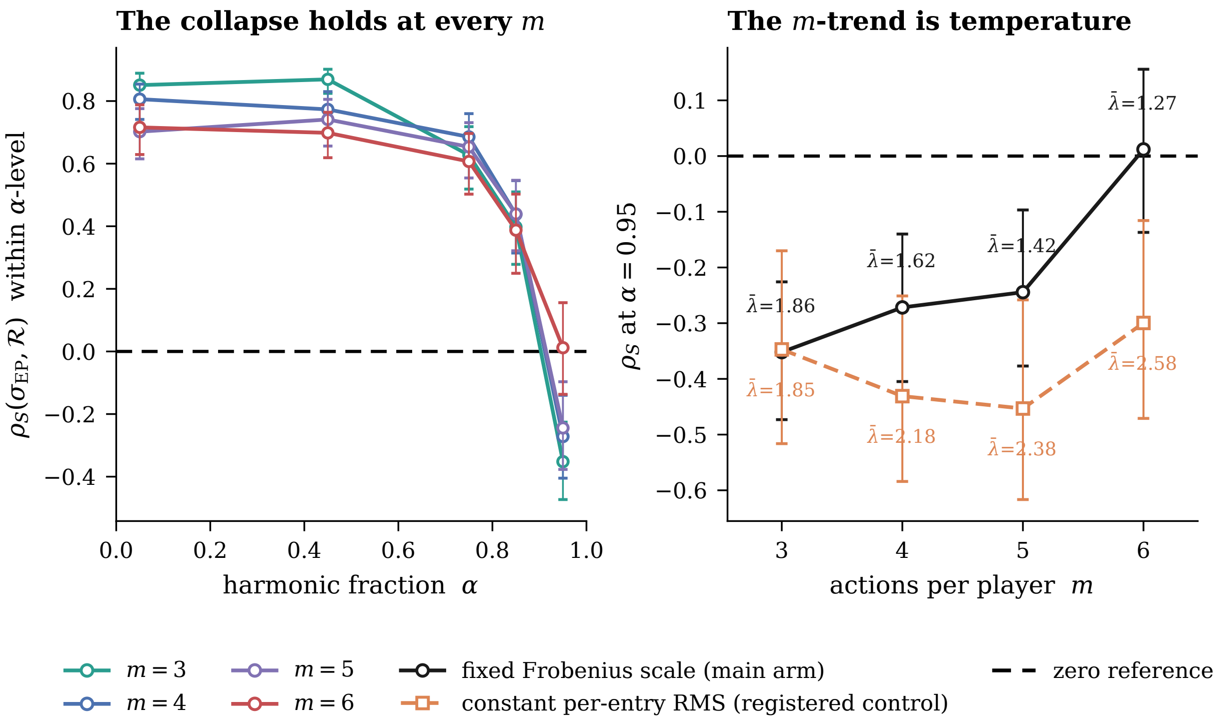

Figure 6. (**a**) Within-level association against harmonic fraction at each action count, with bootstrap intervals and the zero reference. (**b**) The terminal association at the highest swept level under fixed Frobenius payoff scale against the registered constant-per-entry-RMS control, annotated with the median normalised precision each arm actually ran at.

5.9. Calibration

Both meters read zero where zero is provably correct, on data the project did not generate. Sioux Falls road network: $\mathcal{R} = 5.6 \times 10^{-17}$, Fisk stochastic-user-equilibrium KKT spread 7.1×10^{-15}, directional-derivative symmetry defect exactly 0. A degenerate equal-value Blotto instance: $\sigma_{\mathrm{EP}} = 5.4 \times 10^{-32}$. Colonel Blotto at budget 3: $\alpha = 0.694$, $\mathcal{R} = 0.118$; rock–paper–scissors: $\mathcal{R} \approx 0.69$–0.87.

5.10. Hypothesis scoreboard

Table 5. Registered hypotheses and outcomes. Criteria and levels were fixed before the runs.

Hypothesis	**Criterion**	**Result**	**Outcome**
H1 conditional decline	monotone over top 3 levels (m=3)	$0.610 \to 0.323 \to -0.355$	Accepted
H2a ratio artefact	numerator $\rho_S > 0.5$ at high α	-0.368	**Not supported**
H2b denominator-driven	$\rho_S > 0.8$ at high α	0.955	Accepted
H3 collapse universal	degrades at every (λ, m)	holds 3×2 cells	Accepted
H3b sign λ-invariant	same sign across λ	$+0.25/+0.03/-0.23$	**Not supported**
H4 observable on real data	both coordinates, no payoffs	one coordinate each, 2 systems	Partly accepted
Kill-shots, registered before the experiment existed			
H5 finite size, *first* registration	$\Delta\rho_{\mathrm{hi}} > 0.40$ / ≤ 0.20	$+0.364$: neither branch	**Indeterminate**
H5 finite size, *replacement*	gap ratio vs 0.50 bar	0.585 $[+0.434, +0.749]$, crosses bar	Survives on the point est.
H5 as reported	earlier registration binds	registrations disagree	**Indeterminate**
H5 ceiling condition	$\rho_S(m, 0.95)$ CI below 0.35	worst $+0.156$; ends disjoint at every m	**Holds; carries the claim**
H5, the sign half	—	$+0.012$ $[-0.137, +0.156]$, m=6	**Reversal not general**
H5b onset drift	drift ≤ 0.10 in α	0.00 at every m	Passed
H6 solver artefact	$\lvert\Delta\rho_S\rvert < 0.05$, 4 comparisons	0.00000; $\mathcal{R}$ moved $\sim 10^{-12}$	Diagnostic, not a test
D1 payoff-scale control	arms must agree	agree; brackets $\bar{\lambda}$, does not fix it	Confound found, unpaired

6. Discussion

6.1. Why a local object and a global object come apart

χ^{eq} is a derivative evaluated at a single equilibrium; σ_{EP} is a functional of a flux field over the entire profile space. Near $\alpha = 0$ the potential component modulates both, which is why they track: the potential is a scalar field whose gradient governs the local response and whose level sets govern the stationary measure. As the potential component vanishes, nothing remains to force a local object and a global object into agreement, and they stop agreeing. This is not a statement peculiar to games. It predicts that in any system where a local susceptibility and a global flux functional are both used as proxies for irreversibility, the two will diverge precisely where the underlying potential structure does—and it says where to look.

6.2. What $\mathcal{R}$ is not

$\mathcal{R}$ is not a measure of dissipation, and this paper's main practical warning is against reading it as one. It is also not a fraction: it is a ratio of Frobenius norms, unbounded above, and values greater than 1 occur (matching pennies reads 1.2). Its *zero test* is λ-free—$\mathcal{R} = 0$ if and only if the normalised game is potential, at every λ—but its *magnitude* scales with λ, and an earlier version of this programme's own claim that $\mathcal{R}$ was "λ-free" was corrected on exactly this point. Any reported magnitude must carry the λ it was read at.

6.3. Consequences for practice

The plane has four quadrants and they differ in what a decision-maker should do. Reciprocal-and-quiet systems admit comparative statics and static rival models. Reciprocal-but-circulating systems are being driven from outside, so timing matters and structure does not. Asymmetric-but-quiet systems have a structural leader, so pass-through asymmetry is the exploitable object and there is no cycle to time. Systems that are both are where naive optimisation against a static rival model is worst. If distance from equilibrium were a scalar, the two off-diagonal quadrants would be empty; the two real systems measured here are not on the same diagonal.

6.4. Theoretical connections

The zero of the local axis is the strategic instance of Onsager reciprocity, and the zero of the global axis is detailed balance; that they coincide in potential games is the exact content of the Gibbs correspondence. Away from it, the structure resembles the failure of the fluctuation–dissipation theorem in a non-equilibrium steady state—with the important difference that here the quantity governing the failure, α, is itself computable from the game, so the breakdown can be parameterised rather than merely observed.

6.5. Limitations

The finite-size limitation has been tested rather than merely acknowledged, and the outcome is **indeterminate**. Sweeping $m \in \{3, 4, 5, 6\}$ under criteria registered in advance (Section 5.8.), two of the three conditions that would have killed the claim fired; the third—a demand that the collapse lose more than half its size—did not, and the gap ratio deciding it, 0.585, carries a bootstrap interval that crosses its own 0.50 bar. Under the *earlier* of this unit's two registrations, which adjudicates the raw recovery $\Delta\rho_{\text{hi}} = +0.364$ against bars of 0.40 and 0.20, neither branch resolves. We therefore report indeterminate and rest the claim on the ceiling condition instead. What the sweep does establish without qualification is that the collapse itself, its ceiling, and the α at which it begins are present at every action count tested.

A criterion substitution occurred inside this unit, and it is reported here rather than left to be discovered. The kill-shot was first registered as `plane_finite_size.yaml` at 23:02:25 on 2026-08-12; an experiment implementing it was written at 23:04:39, executed at 23:04:49, and run again at 23:17:47. At 23:26:36—after both runs—a second configuration, `plane_robustness.yaml`, was committed for the same unit and the same kill-shot with a different deciding statistic, and the change ran in the direction of survival. Each file individually satisfies the discipline this programme requires (committed before the code implementing it existed; never amended after a result existed); the *unit* does not, because substituting a criterion by writing a new file leaves no diff to catch. Both registrations, both experiments and both resolved configurations are committed, so the property is checkable rather than asserted. Two consequences are carried in the text above: the reported verdict is the earlier registration's, and the deciding statistic of the later one now carries the interval the registration failed to demand of it—an interval that does not certify the branch it was read as certifying. The general lesson is that "the config was committed before the code" is not a sufficient guarantee; the binding property is that a unit has *one* live criterion for one question, and that any replacement is registered as a replacement.

The control arms are unpaired, which is a design defect rather than a caveat. The payoff-scale control D1 and the precision control D2 draw from different seed offsets than the main arm and use half the games per cell, so every arm-to-arm comparison in Section 5.8. carries between-sample noise that a paired design—same games, same order, only the controlled quantity varying—would cancel. The solver check H6 is paired, which is exactly why its bar can be strict. D1 and D2 should be re-run paired, and D1 additionally with $\bar{\lambda}$ fixed by construction rather than bracketed: as run, D1 raises $\bar{\lambda}$ with m where the main arm lowers it, and interpolating between the two arms to matched $\bar{\lambda}$ leaves most of the observed m-effect standing. D2 is a further signal that the gap-ratio criterion is not resolvable at this sample size: at the precision $\lambda = 1.5$ it returns a gap ratio of 0.529, a 2.9-point margin against the same 0.50 bar on 100 games per cell.

The second limitation is dimension in the other index: **everything here is at** $N = 2$. Nothing in this paper speaks to N-scaling, and it is the obvious next kill-shot— the same design, run over N rather than m, with the same three-branch criterion. The binding cost is the dense Glauber generator over $\prod_i m_i$ profiles, not a correctness limit; m stops at 6 (36 joint states) for the same reason.

A third limitation is the sign, and the m-sweep sharpened it. The reversal is one seed family's realisation of an effect whose sign varies with λ and, at fixed Frobenius payoff scale, with m: at $m = 6$ the terminal association is statistically indistinguishable from zero. The paper's claim is therefore decorrelation, not anti-correlation—which is all independence requires—and any use of the reversal must carry the (m, λ) it was read at. Table 5 records H3b as not supported and the reversal half of H5 as not general, rather than presenting either as established.

A fourth limitation is generic to the generator and is a warning to anyone extending this work: because the family unit-normalises each Hodge component and fixes the Frobenius norm, per-entry payoff RMS falls like $1/m$, so a larger action set is also a colder game. **Any future sweep in m on this family must control per-entry payoff RMS, or it will confound temperature with dimension.** The registered control (Table 4) is how the confound was caught here, and it is also why the m-trend cannot be attributed to temperature outright: the control brackets $\bar{\lambda}$ rather than matching it, and at matched $\bar{\lambda}$ a residual m-effect of roughly 0.22 of the observed 0.364 remains. The same caution applies to comparing $\mathcal{R}$ or σ_{EP} magnitudes across action counts at fixed Frobenius scale.

Solver dependence is no longer an open limitation, though the check that closed it had less power than its margin suggests. H6—a structurally different, last-iterate-convergent method and a 100× tighter tolerance, on the identical seed family, criterion $|\Delta\rho_S| < 0.05$—returned 0.00000 in all four comparisons. Because the two arms move $\mathcal{R}$ by only $\sim 10^{-12}$ and ρ_S is rank-based, that outcome was forced rather than earned; what the check genuinely establishes is the diagnostic that the solver contributes $\sim 10^{-12}$ to $\mathcal{R}$ and leaves its rank ordering bit-identical, which is enough to exclude finite-precision path dependence. What remains untested is a bias shared by both methods, which no comparison of two converging solvers can exclude; exact homotopy continuation would be the next distinct probe.

A fifth limitation is that no system has yet been measured on *both* coordinates. The retail panel has $\mathcal{R}$ without σ_{EP}; the electricity market has σ_{EP} without $\mathcal{R}$. Until one system carries both, the plane is demonstrated on synthetic families and anchored, not populated, by real data. This is a real limitation of the present evidence rather than evidence of generality.

A sixth limitation concerns the empirical anchors themselves. The retail panel is a single chain in one metropolitan area over 1989–1997, and its cross-"firm" interpretation is a modelling assumption, not a data feature; the near-symmetric reading is consistent with a single retailer optimising a category, which is what was predicted, but it is not evidence about rival firms.

Finally, prior art. The nearest live work [10] was assessed as orthogonal, but that assessment is inherited from an internal literature audit and has not been independently re-run for this paper. It is registered as an open check.

6.6. Broader relevance

The two coordinates require no knowledge of payoffs, which is what makes them usable outside game theory: any system that emits a sequence of joint actions and permits a perturbation experiment can be located in the plane. Congestion networks, wholesale electricity markets, retail pricing panels, and multi-agent learning systems all satisfy that description, and the first three appear here.

7. Conclusions and Future Work

Response asymmetry and dissipation share a zero and are otherwise independent. Their near-perfect unconditional agreement ($\rho_S = 0.993$) is a confound created by the harmonic fraction driving both; conditioning it away, the association degrades monotonically and *collapses to zero* at high α—$+0.72 \to 0.00$ $[-0.14, +0.16]$ with disjoint intervals at the largest action count tested, and a collapse of the same shape at every $m \in \{3, 4, 5, 6\}$ at $N = 2$—and the natural repair—that the collapse is an artefact of $\mathcal{R}$ being a norm ratio—is refuted by its own pre-registered test (-0.368 for the numerator alone). Decorrelation is what independence requires, and decorrelation is what is claimed. Distance from equilibrium is therefore not a scalar for strategic systems, and the $(\mathcal{R}, \sigma_{\mathrm{EP}})$ plane is the space in which an observed system should be classified.

The claim is bounded deliberately. The collapse is universal across every precision and action count tested; the *sign* of the residual is not—it varies with λ, and at fixed Frobenius payoff scale it varies with m, reaching $+0.012$ $[-0.137, +0.156]$ at $m = 6$—so the reversal is reported with its scope rather than as a result. Two of six registered hypotheses were not supported, and both are reported as such.

Three things follow. First, the two most plausible ways this result is wrong—finite size and solver dependence—were registered as kill-shots and have now been run. The solver one is settled: two structurally different methods move $\mathcal{R}$ by $\sim 10^{-12}$ and leave its rank ordering bit-identical, which excludes finite-precision path dependence, and we report it as that diagnostic rather than as a kill-shot survived with margin, because a rank statistic could not have registered a perturbation that small. The finite-size one is **indeterminate under its own first-registered criterion**: two of its three refutation conditions were met, the third turns on a gap ratio whose interval crosses its bar, and the earlier of this unit's two registrations—which adjudicates the raw recovery $+0.364$ against bars of 0.40 and 0.20—resolves neither branch. The claim therefore rests not on that criterion but on *decorrelation at the ceiling*: the high-α association never recovers at any tested action count, worst upper interval endpoint $+0.156$ against a 0.35 ceiling, terminal intervals disjoint from the low-α ones at every m. The sweep additionally showed that the apparent size dependence is largely, though not wholly, a normalised-precision effect of the game generator rather than a dimensional one. The next kill-shot is $N > 2$, which is untested. Second, one quadrant of the plane contains no measured system; a real system with asymmetric response and no persistent circulation would be the strongest available confirmation, because it is exactly the configuration a scalar theory forbids, and retail fuel markets with asymmetric cost pass-through but no observed cycling are the obvious candidate. Third, the reading should be a single function call on observed data, with its refusals reported as bounds rather than silences; that is the form in which a measurement leaves the laboratory that made it.

Back Matter

Funding: This research received no external funding.
Data Availability Statement: All artifacts, configuration files, gate definitions and the figure script are in the project repository. Figures regenerate with `python papers/p2_plane/make_figures.py` and all numerical results with `make reproduce`. The library is available at https://pypi.org/project/strataq/ (accessed on 12 August 2026). Road-network data are from the Transportation Networks repository (Sioux Falls); electricity data are CAISO day-ahead locational marginal prices at node TH_SP15_GEN-APND; retail scanner data are derived from the Dominick's Finer Foods database (Kilts Center) and redistribution of derived artifacts is subject to CC-BY-NC-4.0.
Acknowledgments: Implementation, literature triage and draft preparation were carried out with AI assistance; all scientific claims, criteria and interpretations are the author's, and every reported number is traceable to a committed artifact listed in Appendix A.
Conflicts of Interest: The author declares no conflict of interest.

Abbreviations

AAFT	amplitude-adjusted Fourier transform (surrogate)
CI	confidence interval
EPR	entropy production rate
FDT	fluctuation–dissipation theorem
KLD	Kullback–Leibler divergence
NESS	non-equilibrium steady state
QRE	quantal response equilibrium
SUE	stochastic user equilibrium
TUR	thermodynamic uncertainty relation

A Provenance of Every Number

Each row names the committed artifact a figure or claim is drawn from. Artifacts carry their metrics, effect sizes with confidence intervals, n with a justification, seed, and a reference to the configuration whose commit was verified landed *before* the run.

Table 6. Claim-to-artifact map.

Claim / figure	**Artifact**	**Seed**
Unconditional $\rho_S = 0.9927$; within-level table; Fig. 2a	`chain_comovement`	20260808
Kill-shots H5/H5b/H6, D1 control; Tables 3, 4; Fig. 6	`plane_robustness`	20260812
$\rho_S(\mathcal{R}, \alpha) = 0.9819$	`reciprocity_alpha_sweep`	20260808
H2a refutation, H2b mechanism; Fig. 2b	`decoupling_mechanism`	20260808
Exact / KLD / TUR sweep; Figs. 3, 5	`estimator_alpha_sweep`	—
$\rho(SB)$ surface; Fig. 4a	`phase_map_surface`	—
$\lambda_c(\alpha)$; Fig. 4b	`frontier_lambda_c`	20260811
Sioux Falls calibration	`sioux_falls_calibration`	20260808
Blotto readings	`blotto_readings`	20260808
Retail $\mathcal{R} = 0.0011$	`pricing_passthrough_R`	20260811
Electricity ≈ 1.07 nats/day	`electricity_irreversibility_dam`	20260811
Potential-game zeros $\mathcal{R} \leq 9 \times 10^{-17}$	`reciprocity_potential`	20260808
χ^{eq} vs finite differences, 1.3×10^{-8}	`chi_fd_agreement`	20260808
Reference-solver agreement, 10^{-8}	`gambit_agreement`	—

Kill-shot registration, and the substitution inside it. The criteria for H5, H5b, H6 and the D1 payoff-scale control were written into `plane_robustness`'s configuration and committed *before* the experiment file implementing them existed; neither that file nor its predecessor was amended after a result existed, and the branch that fires when the design cannot discriminate (*indeterminate*) was registered alongside the pass and fail branches so the run could not be read as confirmation by default. *It is not, however, true that this unit had a single criterion throughout.* H5 was first registered in `plane_finite_size.yaml` (committed 23:02:25 on 2026-08-12), an experiment implementing it was executed at 23:04:49 and again at 23:17:47, and `plane_robustness.yaml` was committed at 23:26:36 with a different deciding statistic for the same kill-shot. The superseded configuration, its experiment and its resolved configuration are committed alongside the artifacts below, so the ordering of registration and execution is checkable from the repository rather than asserted here; the verdict this paper reports for H5 is the earlier registration's. The $m = 3$ cells of that sweep reconstruct the `chain_comovement` seed stream exactly, so the first 100 games of each are the same games: the measured deviation from the published values at $\alpha \in \{0.85, 0.95\}$ is 0.000.

Corrections carried in the record. Five claims made earlier in this programme were withdrawn and the withdrawals are part of the evidence: a reading of the finite-size sweep as a narrow *survival*, corrected to *indeterminate* once the earlier registration of the same criterion was accounted for; a statement that the registered payoff-scale control removes the m-trend entirely, corrected to *largely, not wholly*; an early statement that $\mathcal{R}$ is λ-free (only the zero test is); an intermediate "certified null" verdict on a five-minute electricity series that was retracted when the null class was shown not to bracket the data; and a transcription of the stratified sweep's precision as $\lambda = 1.5$, which the resolved configuration of `chain_comovement` shows to be $\lambda = 1.2$ (Table 2 carries the corrected value, and the robustness sweep was run at 1.2 for that reason). All three are reflected in the text above.

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